Zheng Rong JIA

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zr-jia.github.io | Google Scholar | ORCID | IEEE Xplore
Scopus | Web of Science | ResearchGate | GitHub | LinkedIn

RESEARCH INTERESTS

Clinical Prediction, Trustworthy Clinical AI, Reproducible Pipelines

EDUCATION

Asia Pacific University of Technology & Innovation — Kuala Lumpur, Malaysia
Master of Science in Artificial Intelligence — Aug 2026 – Aug 2027 (expected)

Macau University of Science and Technology — Macau, China
Bachelor of Science in Software Engineering — Graduated: Aug 2025

PUBLICATIONS

*(*Corresponding author)*

[2] **Zheng Rong JIA***, Kwong-Cheong Wong*. "DualTower-FT with an Adaptive Runtime Safeguard: A Deep Tabular Approach for ICU Stroke Mortality." *The 23rd Pacific Rim International Conference on Artificial Intelligence (PRICAI 2026)*, Guangzhou, China, Nov 16–20, 2026. Short paper, accepted. *(CORE Rank B)*

[1] **Zheng Rong JIA***, Kwong-Cheong Wong*. "Deep Learning for Stroke Mortality Prediction in eICU: A Dual-Tower Transformer Framework." *2026 6th International Conference on Computer Communication and Artificial Intelligence (CCAI)*, Nanjing, China, May 22–24, 2026, pp. 76–81. DOI: 10.1109/CCAI69603.2026.11641950.
Awards: Best Industrial Paper Award; Best Presentation Award.

RESEARCH EXPERIENCE

Medical AI & Predictive Modeling — Aug 2025 – Present
Lead Researcher

- Engineered a Dual-Tower Transformer (DT-Transformer) for stroke mortality prediction using the multi-center eICU database, narrowing the performance gap between deep learning architectures and gradient boosting baselines.
- Designed a decoupled architecture that encodes categorical demographics with learnable embeddings and numerical vitals with a multi-head Self-Attention encoder, yielding a 14.4% relative AUPRC gain over the strongest neural baseline.
- Implemented an Adaptive Runtime Safeguard to ensure inference stability against physiological outliers, achieving a stable mean AUPRC of 0.6171 ± 0.0058 over five random seeds and a $3.4\times$ lower cross-seed standard deviation than a standard Transformer.

Macau University of Science and Technology Sep 2024 – May 2025
Undergraduate Researcher (Final Year Project)
Multi-timescale LSTM for daily, weekly and quarterly power load forecasting, with an interactive decision-support system: 1.42% MAPE on daily predictions and 82% higher quarterly accuracy than ARIMA.

SELECTED PROJECTS

Dual-Tower Transformer for eICU Stroke Mortality 2025 – Present

- Open-sourced the reference implementation of the DT-Transformer architecture and evaluation pipeline supporting the CCAI 2026 paper.

Data Preprocessing for eICU 2025 – Present

- Released a reproducible cohort extraction and feature engineering pipeline for the multi-center eICU database.

Academic Portfolio Website (zr-jia.github.io) 2025 – Present

- Developed and deployed a responsive personal academic website using GitHub Pages to aggregate research publications, CV, and open-source contributions.

PROFESSIONAL EXPERIENCE

Asia AI Education and Future Technology Association (AAE) Remote / Hong Kong, China
Young Ambassador and Researcher Aug 2025 – Present

- Collaborated closely with Head Advisor Dr. Kwong-Cheong Wong to conduct research on Deep Learning for Stroke Mortality Prediction using the eICU database.
- Represented the association at CCAI 2026 (Nanjing, China), where the resulting paper received the Best Industrial Paper Award and the Best Presentation Award.

China Southern Power Grid Guangzhou, China
Machine Learning Engineering Intern Jul 2024 – Aug 2024

- Processed 60GB of multi-province power equipment inspection data to identify equipment fault frequencies and optimize inspection robot algorithms.
- Contributed to the development of closing and splitting functions for power inspection robots to support real-world operational accuracy.

ACADEMIC SERVICE

Pacific Rim International Conference on Artificial Intelligence (PRICAI) 2026
Reviewer and Program Committee Member

- Reviewed four submissions and was subsequently invited to serve as a Program Committee member.